<u>So far</u>: Methods for proving a sequence converges:

⓪ Definition

① Limit laws

② Monotone + bounded $\Rightarrow$ convergent

<u>Ex</u> Prove that the sequence

$$a_1 = 3, \qquad a_{n+1} = \frac{a_n}{2} + \frac{3}{2a_n} \quad \forall n \geq 1$$

converges, and find its limit.

n:	1	2	3	4
a_n:	3	2	$\frac{7}{4}$	$\frac{97}{56} \approx 1.73214$
				$\sqrt{3} \approx 1.73205$

<u>Pf</u>: Claim: $\{a_n\}_{n\geq 1}$ is decreasing and bounded.
We will prove

$$\sqrt{3} < a_{n+1} < a_n \leq 3 \quad \forall n \geq 1$$

by induction.

- Base case: For $n=1$, we have "$\sqrt{3} < 2 < 3 \leq 3$", which is true.
- Inductive step: Suppose $\sqrt{3} < a_{n+1} < a_n \leq 3$.
  Want $\sqrt{3} < a_{n+2} < a_{n+1} \leq 3$:

$$a_{n+2} = \frac{a_{n+1}}{2} + \frac{3}{2a_{n+1}} < a_{n+1} \iff a_{n+1}^2 + 3 < 2a_{n+1}^2$$

$$\iff 3 < a_{n+1}^2,$$

which is true

$$a_{n+2} = \frac{a_{n+1}}{2} + \frac{3}{2a_{n+1}} > \sqrt{3} \iff a_{n+1}^2 + 3 > 2\sqrt{3}\,a_{n+1}$$

$$\iff \underbrace{a_{n+1}^2 - 2\sqrt{3}\,a_{n+1} + 3}_{= (a_{n+1} - \sqrt{3})^2} > 0$$

which is true

By induction, we conclude $\sqrt{3} < a_{n+1} < a_n \leq 3 \quad \forall n \geq 1$.

As the sequence $\{a_n\}_{n\geq 1}$ is decreasing and bounded, then it must converge to some $a \in \mathbb{R}$ (by the previous theorem). Note that $a \neq 0$, since $a_{n+1} > \sqrt{3} \quad \forall n \geq 1 \Rightarrow a \geq \sqrt{3}$. So, by the limit laws,

$$a = \lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} \left(\frac{a_n}{2} + \frac{3}{2a_n} \right) = \frac{a}{2} + \frac{3}{2a}$$

$$\Rightarrow 2a^2 = a^2 + 3 \Rightarrow a^2 = 3$$

$$\Rightarrow a = \sqrt{3} \quad \text{or} \quad \cancel{a = -\sqrt{3}}. \quad \square$$

- Here, we needed to know what the value of limit was beforehand.
- What if we don't know the answer? E.g., in calculus we often define e to be $\lim_{n\to\infty} (1+\frac{1}{n})^n$
- Here is a criterion that does not involve knowing the limit beforehand:

<u>Def</u> We say that $\{a_n\}_{n\geq 1}$ is a <u>Cauchy</u> sequence

if: $\forall \varepsilon > 0 \ \exists N \in \mathbb{N}$ s.t.

$$|a_n - a_m| < \varepsilon \quad \forall n, m \geq N.$$

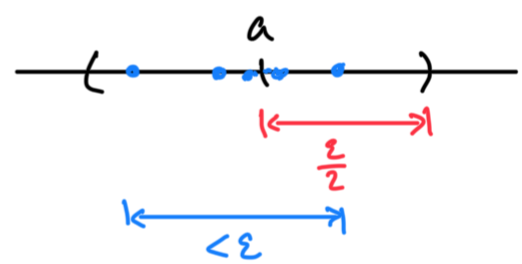
<u>Thm</u> (Cauchy criterion) Let $\{a_n\}_{n\geq 1}$ be a sequence. The sequence $\{a_n\}_{n\geq 1}$ is Cauchy if and only if it converges.

- We will split the proof up into steps
- First, let's prove "$\Leftarrow$":

<u>Prop</u> If $\{a_n\}_{n\geq 1}$ converges, then it is Cauchy.

<u>Pf</u>: Suppose $\{a_n\}_{n\geq 1}$ converges to $a \in \mathbb{R}$. Fix $\varepsilon > 0$. As $\lim_{n\to\infty} a_n = a$, then $\exists N \in \mathbb{N}$ s.t.

$$|a_n - a| < \frac{\varepsilon}{2} \quad \forall n \geq N.$$



Then, for $m, n \geq N$ we have

$$|a_n - a_m| \leq |a_n - a| + |a - a_m|$$
$$< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

So $\{a_n\}_{n\geq 1}$ is a Cauchy sequence. $\square$

- Next, we turn to "$\Rightarrow$"
- First step:

<u>Lem</u> If $\{a_n\}_{n\geq 1}$ is a Cauchy sequence, then it is bounded.

<u>Pf</u>: Suppose $\{a_n\}_{n\geq 1}$ is Cauchy. Taking $\varepsilon = 10$

in the definition of Cauchy, we get: $\exists N \in \mathbb{N}$ s.t.

$$|a_n - a_m| < 10 \quad \forall m, n \geq N.$$

In particular,

$$|a_n| \leq |a_N| + |a_n - a_N| < |a_N| + 10 \quad \forall n \geq N$$

Set $M = \max\{|a_N| + 10, |a_1|, |a_2|, \ldots, |a_{N-1}|\}$. Then

$$|a_n| \leq M \quad \forall n \geq 1,$$

so $\{a_n\}_{n \geq 1}$ is bounded. $\square$

- Step 1 of $\boxed{\Rightarrow}$: Cauchy $\Rightarrow$ bounded
- Step 2 will be: Bounded $\Rightarrow$ $\exists$ convergent subsequence

<u>Def</u> Let $\{a_n\}_{n \geq 1}$ be a sequence. We say $\{a_{k_n}\}_{n \geq 1}$ is a <u>subsequence</u> of $\{a_n\}_{n \geq 1}$ if $\{k_n\}_{n \geq 1}$ is a strictly increasing sequence of natural numbers:

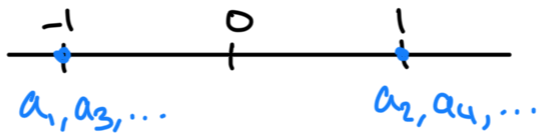
$$k_n \in \mathbb{N} \quad \text{and} \quad k_n < k_{n+1} \quad \forall n \in \mathbb{N}.$$

<u>Rmk</u> Any such $\{k_n\}_{n \geq 1}$ must satisfy $k_n \geq n$ $\forall n \geq 1$. This is easy to prove by induction. (Exercise.)

<u>Ex</u> ① $\{a_{n^2}\}_{n \geq 1}$ is a subsequence of $\{a_n\}_{n \geq 1}$

② $\{a_{p_n}\}_{n \geq 1}$, where p_n denotes the nth prime number, is a subsequence of $\{a_n\}_{n \geq 1}$.

③ The sequence $\{a_n\}_{n \geq 1}$ with $a_n = (-1)^n$ is bounded, but diverges. However, the subsequence $\{a_{2n}\}_{n \geq 1}$ converges to 1, since $a_{2n} = 1$ $\forall n \geq 1$.



<u>Thm</u> Let $\{a_n\}_{n\geq 1}$ be a sequence and $a \in \mathbb{R}$. The sequence $\{a_n\}_{n\geq 1}$ converges to a if and only if every subsequence $\{a_{k_n}\}_{n\geq 1}$ converges to a.

<u>Pf</u>: <u>$\Leftarrow$</u>|: Take the subsequence $k_n = n$ $\forall n \geq 1$.

<u>$\Rightarrow$</u>|: Suppose $\lim_{n\to\infty} a_n = a$ and $\{a_{k_n}\}_{n\geq 1}$ is a subsequence of $\{a_n\}_{n\geq 1}$. Want: $\lim_{n\to\infty} a_{k_n} = a$. Fix $\varepsilon > 0$. As $\lim_{n\to\infty} a_n = a$, then $\exists N \in \mathbb{N}$ s.t.

$$|a_n - a| < \varepsilon \quad \forall n \geq N.$$

Recall that $k_n \geq n$ $\forall n \geq 1$. So for $n \geq N$ we have $k_n \geq n \geq N$, and so

$$|a_{k_n} - a| < \varepsilon \quad \forall n \geq N.$$

Therefore $\lim_{n\to\infty} a_{k_n} = a$. $\square$